

Supplemental Materials

This document provides supplemental materials for the manuscript entitled “*Poriferous Surface Coverage of Networked Euler-Lagrange Systems via Poly-Annulus Conformal Mapping*” by Xun Feng, Chao Zhai, Chuan-Ke Zhang, and Yong He. It is composed of three parts. The first part presents some theoretical results, including key definitions and the derivation process of the mapping. The second part provides simulation results on the poriferous surface and relevant discussions, which include the analysis of computational cost, stability, and robustness. The third part discusses the changes required in the theoretical proof when the dynamics are reduced to a single integrator. It is worth noting that the notations for theorems, lemmas, propositions, and citations in the supplementary material are consistent with those in the main text.

S-I Theoretical Results

S-I.1 Formulation of Euler-Lagrange Dynamics

In standard Euclidean space, the dynamics of a fully actuated point mass are governed by Newton’s second law $\ddot{p}_i = u_i$. However, when the agent navigates a curved Riemannian manifold (S, η) parameterized by local coordinates $\mathbf{q} = [q^1, q^2]^T$, the non-Euclidean metric introduces velocity-dependent fictitious forces. For a purely mechanical system without a conservative potential field (where the artificial potential is instead handled by the control input u_i), the Lagrange L consists entirely of the kinetic energy

$$L(\mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \langle \dot{\mathbf{q}}, \dot{\mathbf{q}} \rangle_\eta = \frac{1}{2} \sum_{j,k} \eta_{jk}(\mathbf{q}) \dot{q}^j \dot{q}^k,$$

where η_{jk} are the components of the Riemannian metric η . The motion of the agent is governed by the classical Euler-Lagrange equations:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^m} \right) - \frac{\partial L}{\partial q^m} = u_m, \quad m \in \{1, 2\},$$

where u_m represents the covariant components of the control input u_i . Evaluating the partial derivatives of the Lagrange

$$\begin{aligned} \frac{\partial L}{\partial \dot{q}^m} &= \sum_k \eta_{mk} \dot{q}^k, \\ \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^m} \right) &= \sum_k \eta_{mk} \ddot{q}^k + \sum_{j,k} \frac{\partial \eta_{mk}}{\partial q^j} \dot{q}^j \dot{q}^k, \\ \frac{\partial L}{\partial q^m} &= \frac{1}{2} \sum_{j,k} \frac{\partial \eta_{jk}}{\partial q^m} \dot{q}^j \dot{q}^k. \end{aligned}$$

Substituting these derivatives back into the Euler-Lagrange equation and multiplying by the inverse metric η^{lm} isolates

the linear acceleration $\ddot{q}^l$

$$\ddot{q}^l + \sum_{m,j,k} \frac{1}{2} \eta^{lm} \left(\frac{\partial \eta_{mk}}{\partial q^j} + \frac{\partial \eta_{mj}}{\partial q^k} - \frac{\partial \eta_{jk}}{\partial q^m} \right) \dot{q}^j \dot{q}^k = u^l,$$

where $u^l = \sum_m \eta^{lm} u_m$ are the contravariant components of the control input. The bracketed term multiplied by the inverse metric is exactly the definition of the Christoffel symbols of the second kind, Γ_{jk}^l . Therefore, the expanded dynamic equation becomes

$$\ddot{q}^l + \sum_{j,k} \Gamma_{jk}^l \dot{q}^j \dot{q}^k = u^l.$$

This expanded form explicitly reveals the physical components of the motion:

- (1) $\ddot{q}^l$: The linear acceleration representing inertia.
- (2) $\Gamma_{jk}^l \dot{q}^j \dot{q}^k$ ($j \neq k$): The Coriolis forces arising from the coupling of orthogonal velocities on a curved surface.
- (3) $\Gamma_{jj}^l (\dot{q}^j)^2$: The centrifugal forces generated by the curvature of the manifold itself.

The velocity vector $\dot{\mathbf{p}}$ can be expressed as a linear combination of the local tangent basis vectors $\dot{\mathbf{p}} = \sum_k \dot{q}^k \frac{\partial}{\partial q^k}$. The Levi-Civita connection ∇ acts as the intrinsic directional derivative on the manifold. To evaluate the acceleration $\nabla_{\dot{\mathbf{p}}} \dot{\mathbf{p}}$, we substitute the expanded velocity vector into the operator

$$\begin{aligned} \nabla_{\dot{\mathbf{p}}} \dot{\mathbf{p}} &= \nabla_{\dot{\mathbf{p}}} \left(\sum_k \dot{q}^k \frac{\partial}{\partial q^k} \right) \\ &= \sum_k \dot{\mathbf{p}}(\dot{q}^k) \frac{\partial}{\partial q^k} + \sum_k \dot{q}^k \nabla_{\dot{\mathbf{p}}} \left(\frac{\partial}{\partial q^k} \right) \\ &= \sum_k \ddot{q}^k \frac{\partial}{\partial q^k} + \sum_{j,k} \dot{q}^j \dot{q}^k \nabla_{\frac{\partial}{\partial q^j}} \left(\frac{\partial}{\partial q^k} \right) \\ &= \sum_l \left(\ddot{q}^l + \sum_{j,k} \Gamma_{jk}^l \dot{q}^j \dot{q}^k \right) \frac{\partial}{\partial q^l}, \end{aligned}$$

where Γ_{jk}^l is the Christoffel symbols of the second kind (It measures exactly how much the k -th basis vector tilts toward the l -th basis vector as one moves in the j -th direction). Since the control input u_i acting on the agent can be decomposed into its contravariant components u^l : $u_i = \sum_l u^l \frac{\partial}{\partial q^l}$, we obtain

$$\nabla_{\dot{\mathbf{p}}} \dot{\mathbf{p}} = u_i.$$

Thus, the equation characterizes a second-order mechanical system subject to inertia, Coriolis, and centrifugal forces induced by the complex topology of the poriferous workspace.

S-I.2 Computational Complexity of Partition Operator $\mathcal{P}$

The topological decomposition of the workspace S via the operator $\mathcal{P}$ is executed on a discrete triangular mesh, which

is intrinsically modeled as an undirected graph $\mathcal{G} = (V, E)$. To efficiently construct the proximity partition, the algorithm initializes a min-heap priority queue with the vertices of the n inner boundaries ∂O_k . A multi-source Dijkstra algorithm is deployed to propagate the discrete geodesic distance field globally in a single traversal, incurring a time complexity of $\mathcal{O}(|E| \log |V|)$ [Choi, 2021]. Subsequently, to execute the branch cuts γ_k that annihilate local homology generators, localized shortest-path searches are performed within each disjoint proximity subregion (V_k, E_k) . Because these subregions are mutually exclusive ($\sum_{k=1}^n |V_k| \leq |V|$ and $\sum_{k=1}^n |E_k| \leq |E|$), the aggregated complexity for all necessary branch cuts is strictly bounded by $\sum_{k=1}^n \mathcal{O}(|E_k| \log |V_k|) \leq \mathcal{O}(|E| \log |V|)$. Crucially, since the 2-D manifold S is discretized as a triangular surface mesh, its topological structure is governed by the Euler characteristic $\chi(S) = |V| - |E| + |F| = 1 - n$. Given that each triangular face is bounded by three edges ($3|F| \leq 2|E|$), the number of edges inherently satisfies the sparsity constraint $|E| \leq 3|V| - 3(1 - n)$, which establishes the asymptotic structural bound $|E| = \mathcal{O}(|V|)$. Substituting this sparsity property into the sequential graph traversals, the overall computational complexity of the partition operator $\mathcal{P}$ collapses to

$$\mathcal{O}(|V| \log |V|).$$

By replacing the iterative numerical approximations of continuous Eikonal PDE solvers with this discrete graph-theoretic formulation [Kumar *et al.*, 2010], the operator achieves exact topological decomposition without numerical diffusion, scaling efficiently with the mesh resolution.

S-I.3 Partial Welding Example

To illustrate the mathematical mechanism of partial welding, we present an intuitive example. Reconstruct the topological connectivity of a slit domain $\mathcal{D} = \mathbb{C} \setminus [-i, i]$ by welding its separated boundaries. Let the slit $[-i, i]$ be regarded as two distinct boundary segments: $\gamma^+ = \{iy + \varepsilon \mid y \in [-1, 1]\}$, and $\gamma^- = \{iy - \varepsilon \mid y \in [-1, 1]\}$, where $\varepsilon \rightarrow 0^+$. The welding mapping $T: \mathcal{D} \rightarrow \Omega$ is defined by the elementary function $w = T(z) = \sqrt{z^2 + 1}$. This mapping transforms the slit domain $\mathcal{D}$ into the $\Omega = \mathbb{C} \setminus (-\infty, 0]$. Crucially, it enforces the boundary identification condition $T(iy^+) = T(iy^-) = \sqrt{1 - y^2}$, $\forall y \in [-1, 1]$. This implies that the disjoint banks γ^+ and γ^- are mapped to the exact same locus in Ω , thereby welding the cut back together. In our proposed framework, the optimization of Θ_{uv} serves as the numerical inverse operator of such welding. It identifies the boundary correspondence $h_u(\partial S_u) \cong \Theta_{uv}(h_v(\partial S_v))$, which restores the continuous topology of poly-annulus S from the decomposed disks.

S-I.4 Geometric Computation of Safe Boundary

This appendix details the numerical generation of the partition bar $\hat{\Gamma}_i$ in Ξ , consisting of linear segments and circular arcs. For intersection identification, the i -th nominal partition bar is parameterized as a vector $\hat{\mathbf{x}}(\hat{\xi}) = \hat{\xi} \hat{\mathbf{b}}_i$, where $\hat{\xi} \in [0, 1]$. The k -th obstacle is defined by its center $\hat{o}_k$ and radius $\hat{r}_k$.

The intersection points are determined by $\|\hat{\mathbf{x}}(\hat{\xi}) - \hat{o}_k\|^2 = \hat{r}_k^2$, which yields the quadratic equation

$$|\hat{\mathbf{b}}_i|^2 \hat{\xi}^2 - 2(\hat{\mathbf{b}}_i \cdot \hat{o}_k) \hat{\xi} + (|\hat{o}_k|^2 - \hat{r}_k^2) = 0.$$

Solving for $\hat{\xi}$ gives the entry and exit parameters $\hat{\xi}_{in} < \hat{\xi}_{out}$, with corresponding coordinates $\hat{p}_{in} = \hat{\xi}_{in} \hat{\mathbf{b}}_i$ and $\hat{p}_{out} = \hat{\xi}_{out} \hat{\mathbf{b}}_i$. To bypass the obstacle, a circular arc is generated on the buffer circle with radius $\hat{r}_{k,i} = \hat{r}_k(1 + i\beta)$. The entry and exit angles relative to $\hat{o}_k$ are computed as $\hat{\psi}_{in} = \text{atan}(\hat{p}_{in} - \hat{o}_k)$ and $\hat{\psi}_{out} = \text{atan}(\hat{p}_{out} - \hat{o}_k)$. The arc $\hat{\Gamma}_{k,i}$ is discretized via

$$\hat{\mathbf{x}}_{arc}(u) = \hat{o}_k + \hat{r}_{k,i} \begin{bmatrix} \cos(\hat{\psi}_{in} + u \frac{\Delta \hat{\psi}}{U-1}) \\ \sin(\hat{\psi}_{in} + u \frac{\Delta \hat{\psi}}{U-1}) \end{bmatrix},$$

where $u \in \{0, \dots, U-1\}$. This ensures $\hat{\Gamma}_i$ is piecewise smooth and strictly disjoint from obstacles.

S-I.5 Spectral Properties of $\mathcal{L}_g(\hat{\omega})$

For the weighted Laplacian $\mathcal{L}_g(\hat{\omega})$ in equation (21), under $\hat{\omega}_k \geq \underline{\omega} > 0, \forall k \in \mathcal{N}$, the properties hold

- (1) $\mathcal{L}_g = \mathcal{L}_g^T$ and $\mathcal{L}_g \succeq 0$.
- (2) $\ker(\mathcal{L}_g) = \text{span}(\mathbf{1}_N)$, implying $\sum_{i=1}^N \dot{m}_i = 0$.
- (3) $\lambda_2(\mathcal{L}_g) \geq \underline{\omega} \lambda_2(\mathcal{L}_S) = 2\underline{\omega}(1 - \cos(2\pi/N)) > 0$.

Proof. Positive Semi-definiteness: The symmetry follows from the undirected adjacency in $\mathcal{G}_N$. For any $x \in \mathbb{R}^N$, the quadratic form is $x^T \mathcal{L}_g x = \sum_{k=1}^N \hat{\omega}_k (x_k - x_{k-1})^2 \geq 0$, where $x_0 \equiv x_N$. Then, $\mathcal{L}_g \succeq 0$. Null Space: $x^T \mathcal{L}_g x = 0 \iff x_k = x_{k-1}$ for all k . Hence, $\ker(\mathcal{L}_g) = \text{span}(\mathbf{1}_N)$. Connectivity: Since $\underline{\omega} > 0$, the cycle graph $\mathcal{G}_N$ is connected. By the Courant-Fischer Theorem [Horn *et al.*, 2013, Th. 4.2.2], the second smallest eigenvalue satisfies $\lambda_2(\mathcal{L}_g) \geq \underline{\omega} \lambda_2(\mathcal{L}_S) > 0$, which establishes the exponential stability on $\mathbf{1}_N^\perp$. $\square$

S-I.6 Exponential Convergence Rate

The system Jacobian evaluated at the equilibrium origin $(\hat{\psi}^*, \mathbf{p}^*, \mathbf{0})$ exhibits a strict block lower-triangular structure

$$J_{sys} = \begin{bmatrix} \frac{\partial \hat{\psi}}{\partial \hat{\psi}} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \\ \frac{\partial \mathbf{p}}{\partial \hat{\psi}} & \frac{\partial \mathbf{p}}{\partial \mathbf{p}} & \frac{\partial \mathbf{p}}{\partial \mathbf{p}} \end{bmatrix}.$$

Based on Theorem 4.1, the stability of the MAS is dictated by the bottom-right $2N \times 2N$ block under the fixed optimal partition $\hat{\psi}^*$. Then, for the exponential convergence rate, we consider the coefficient matrix

$$\begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -k_p \nabla^2 J_i(\mathbf{p}^*) & -k_d \mathbf{I} \end{bmatrix}.$$

Using the determinant identity for partitioned block matrices (applicable because the scalar matrices commute with the Hessian block), this reduces to

$$\det(\lambda(\lambda + k_d)\mathbf{I} + k_p \nabla^2 J_i(\mathbf{p}^*)) = 0.$$

Let σ_j denote the j -th eigenvalue of $\nabla^2 J_i(\mathbf{p}^*)$. Since $\nabla^2 J_i \succ 0$, it holds $\sigma_j > 0$. The matrix determinant evaluates to zero if and only if $\lambda^2 + k_d \lambda + k_p \sigma_j = 0$ holds for every eigenvalue σ_j . Therefore, the spectra are $\lambda_j = (-k_d \pm \sqrt{k_d^2 - 4k_p \sigma_j})/2$. Since k_p, k_d , and $\sigma_j > 0$, the discriminant produces either real negative roots (when $k_d^2 \geq 4k_p \sigma_j$) or complex conjugate roots with a strictly negative real part (when $k_d^2 < 4k_p \sigma_j$). In all scenarios, the real part is bounded away from zero in the open left-half plane $\text{Re}(\lambda_j) \leq -\min(k_d, 4k_p \sigma_{\min}/k_d)/2 < 0$. We define $\alpha := \min(k_d, 4k_p \sigma_{\min}/k_d)/2$. The perturbation state $\mathbf{z}(t)$ decays to the origin satisfying the trajectory bound $\|\mathbf{z}(t)\| \leq C e^{-\alpha t} \|\mathbf{z}(0)\|$ for some constant $C > 0$, establishing the local exponential stability at the optimal centroids.

S-I.7 Topological Universality

For a compact, orientable Riemannian manifold $\mathcal{M}$ with $g(\mathcal{M}) = 0$ and $n+1$ boundary components, the De Rham cohomology group is $H_{dR}^1(\mathcal{M}) \cong \mathbb{R}^n$. By the Hodge Decomposition Theorem [Gu *et al.*, 2008], there exists a unique harmonic 1-form ω_h satisfying $\Delta \omega_h = 0$. The conformal mapping $\tau : \mathcal{M} \rightarrow \Xi$ is established via the holomorphic integration of ω_h , uniquely defining the angular coordinate $\hat{\psi}$ as the period integral of the pull-back form $\tau^* \hat{\omega}$. The gradient vector fields $X = -\text{grad}_\eta J$ on $\mathcal{M}$ and $X_\Xi = -\text{grad}_{\hat{\eta}} \hat{J}$ on Ξ represent the geometric restoring forces of the Euler-Lagrange dynamics. They are related via the push-forward operator τ_* as $X_\Xi = \lambda^{-2} \tau_* X$, where λ^2 is the conformal factor. By the Poincaré-Hopf Theorem [Gu *et al.*, 2008], the sum of vector field indices satisfies $\chi(\mathcal{M}) = 1 - n = \chi(\Xi)$. Because the mechanical equilibria of the Euler-Lagrange system strictly coincide with the roots of these gradient fields ($X = 0$), this topological conjugacy inherently preserves the unique equilibrium set, ensuring asymptotic stability on $\mathcal{M}$. If $g(\mathcal{M}) > 0$, the non-commutative fundamental group $\pi_1(\mathcal{M})$ introduces handle cycles, yielding $\dim(H_{dR}^1(\mathcal{M})) > n$. A 1-D angular phase $\hat{\psi} \in [0, 2\pi)$ is topologically insufficient to globally parameterize these handle generators. Thus, $g(\mathcal{M}) = 0$ bounds the topological class for 1-D sectorial coverage.

S-I.8 Theoretical Summary

The theoretical analysis constitutes a rigorous chain of logical deductions, which guarantees the system's reliability across geometric, dynamic, and computational implementations. First, Proposition 3.1 establishes the topological conjugacy between the poly-annulus S and the n -holed disk Ξ via the conformal mapping τ . Building on this equivalent space, Proposition 4.1 and Lemma 4.1 confirm the strict

topological safety of the modified sectorial partitions, ensuring collision avoidance while rigidly bounding the workload deviation induced by the sequence factor β . Regarding control feasibility, Lemma 4.2 proves the regularity of the pull-back Riemannian metric, which asymptotically recovers a static structure. Governed by this metric, Proposition 4.2 guarantees the non-degeneracy and forward invariance of the feasible set $\mathcal{F}_i$, rendering the control input globally well-defined. Subsequently, Proposition 4.3 ensures the existence and reachability of the optimal coverage configurations.

The stability analysis then reveals a robust cascaded convergence structure. Specifically, Theorem 4.1 rigorously proves that the virtual partition dynamics exhibit exponential input-to-state stability (ISS) against geometric perturbations. Coordinated with these robust partitions, Theorem 4.2 guarantees that the multi-agent system asymptotically converges to the optimal Riemannian centroids with local exponential rates. Furthermore, Theorem 4.3 bridges the gap between continuous theory and discrete implementation, validating that the numerical outputs strictly approximate the theoretical optimum within an arbitrarily small tolerance regulated by the precision parameters $(\Delta t, K^*, \beta)$.

S-II Simulations

This section presents the selection of algorithm parameters and provides an analysis of the proposed framework, covering computational complexity, scalability, and robustness.

S-II.1 Parameter Selection and Tuning Guidelines

The successful execution of the coverage framework requires carefully balancing the practical trade-offs associated with specific geometric and dynamic parameters. First, for the spatial resolution, increasing the quantization parameter K^* tightens the phase approximation and reduces the quantization error, but directly increases the algorithmic computational burden. Second, increasing the sequence factor β expands the topological safety buffer, ensuring strict collision avoidance and boundary disjointness; however, an overly conservative (large) β reduces the navigable workspace and increases the theoretical optimality gap of the terminal coverage. For the partition dynamics, selecting a larger partition gain k_ψ accelerates the exponential decay of workload imbalances across the network, though excessively high values risk numerical instability or overshoot in the discrete-time consensus updates. Simultaneously, the motion of the Euler-Lagrange agents is governed by the proportional gain k_p and the derivative gain k_d , which must be jointly tuned. Increasing k_p strengthens the geometric restoring force for faster spatial convergence toward the Riemannian centroids, but risks actuator saturation and severe transient overshoot. To counteract this, k_d must be sufficiently increased to actively dissipate kinetic energy and suppress inertial oscillations; however, an excessively large k_d will render the system heavily overdamped, resulting in sluggish navigation.

Finally, decreasing the convergence threshold T_e rigorously enforces a higher steady-state precision for the workload distribution, albeit at the expense of requiring more iterative control cycles before mission termination.

S-II.2 Computational Complexity

This subsection demonstrates the computational complexity of the proposed framework.

1) Conformal Mapping Phase (Algorithm 1):

Executed once prior to the mission, this phase establishes the topological foundation of the surface. The system first identifies the n obstacle boundaries to establish the safety sequence factor β . Subsequently, it computes the conformal map τ on S . Distributedly solving the associated Laplace equations over a mesh with $|V|$ vertices across M subdomains incurs a one-time computational complexity of $\mathcal{O}((|V| \cdot M^{-1})^{\frac{3}{2}})$.

2) Geometric Partitioning (Algorithm 2):

Evaluating the safe sectorial boundaries requires linear intersection checks between the 1-D partition bars and the n obstacles in Ξ , yielding a computational complexity of $\mathcal{O}(n)$.

3) Online Iterative Control Phase (Algorithm 3):

Executed at every time step Δt during the active mission. The complexity is evaluated per agent per control cycle. Updating the partition bar $\hat{\psi}_i$ via the 1-D discrete Laplacian flow requires each agent to exchange workload data $\hat{m}$ with its two topologically adjacent neighbors ($i-1$ and $i+1$). Consequently, the communication and computation burden per agent per cycle is strictly $\mathcal{O}(1)$. Calculating the pull-back metric η_i , the Christoffel symbols Γ , and the Riemannian gradient $\text{grad}_{\eta} J_i$ requires executing local matrix operations proportional to $\mathcal{O}(d^3)$. Since the manifold dimension is $d=2$, this constitutes a constant-time $\mathcal{O}(1)$ operation.

In summary, although the conformal mapping carries a heavier one-time theoretical cost of $\mathcal{O}((|V| \cdot M^{-1})^{\frac{3}{2}})$, practical implementation verifies it executes in merely 0.875 s. Once initialized, the iterative computational burden per agent per control cycle gracefully collapses to a highly efficient $\mathcal{O}(n)$. As shown in Table 1, the average execution time during the iterative phase is strictly $t_{\text{avg}} \approx 0.025$ s per step. This confirms that the online geometric and mechanical calculations intrinsically support high-frequency (~ 40 Hz) real-time deployment for Euler-Lagrange multi-agent systems.

S-II.3 Scalability Analysis

The poriferous geometry of S imposes a theoretical limit on the agent population N to maintain topological disjointness and input-to-state stability. Given a sequence factor β , the maximal agent capacity $N_{\max}$ is determined by

$$N_{\max} \leq \min \left\{ \frac{(\hat{r}_{\max} \beta)^{-1} \cdot \hat{d}_{\min}}{2}, \gamma \epsilon_{\max} \cdot (C_{\delta} \beta)^{-1} \right\},$$

where $\hat{d}_{\min}$ is the minimum obstacle separation, $\hat{r}_{\max}$ is the maximum obstacle radius, and $\epsilon_{\max}$ is the allowable steady-state workload error. The first term ensures the strict non-overlapping of the mapped buffer zones ($\hat{r}_k(1+N\beta) - \hat{r}_k \leq 0.5\hat{d}_{\min}$), preventing collision pathways. The second term guarantees the ultimate workload error $\|\mathbf{e}(\infty)\| \leq C_{\delta} \beta \cdot \gamma^{-1}$ remains bounded within $\epsilon_{\max}$, where C_{δ} is the geometric sensitivity constant defined in Lemma 4.1. This bound analytically defines the fundamental trade-off between the coverage precision β and the safe scalability of the swarm on the poly-annulus S .

S-II.4 Robustness Analysis

Consider a hardware contingency at $t = T_f$ where a subset of agents $\mathcal{N}_{\mathcal{A}}$ ceases operation, resulting in an active set $\mathcal{A}$ with size $N_{\mathcal{A}} = N - N_{\mathcal{A}}$. The new equilibrium workload becomes $\hat{m}_{\mathcal{A}} = M_{\text{total}} \cdot N_{\mathcal{A}}^{-1}$. Define the error vector $\mathbf{e} \in \mathbb{R}^{N_{\mathcal{A}}}$ with components $e_i = \hat{m}_i - \hat{m}_{\mathcal{A}}$, satisfying $\mathbf{e} \perp \mathbf{1}_{N_{\mathcal{A}}}$. Under the reduced topology, the error dynamics in Ξ follow the discrete Laplacian flow $\dot{\mathbf{e}} = -k_{\hat{\psi}} \mathcal{L}_g(\hat{\omega}) \mathbf{e}$, where $\mathcal{L}_g(\hat{\omega})$ is the weighted Laplacian matrix. For $V = \|\mathbf{e}\|^2/2$, its derivative satisfies

$$\dot{V} = -k_{\hat{\psi}} \mathbf{e}^T \mathcal{L}_g(\hat{\omega}) \mathbf{e} \leq -k_{\hat{\psi}} \cdot \underline{\omega} \cdot \lambda_2(\mathcal{L}_{S, N_{\mathcal{A}}}) \cdot \|\mathbf{e}\|^2,$$

where $\underline{\omega} > 0$ is the minimum marginal density. The exponential convergence rate is $\gamma_{\mathcal{A}} = 2k_{\hat{\psi}} \cdot \underline{\omega} \cdot (1 - \cos(2\pi \cdot N_{\mathcal{A}}^{-1}))$. Since $N_{\mathcal{A}} < N$, the spectral property of the cycle graph inherently ensures $\gamma_{\mathcal{A}} > \gamma_{\mathcal{N}}$. The re-balancing speed automatically accelerates as the swarm size reduces, ensuring that the error $\|\mathbf{e}(t)\| \leq \|\mathbf{e}(T_f)\| \exp(-\gamma_{\mathcal{A}}(t - T_f))$ swiftly decays. Crucially, because the partition dynamics are strictly decoupled from the inertia of the agents, this intrinsic self-healing property allows the remaining Euler-Lagrange agents to smoothly track the shifting Riemannian centroids without triggering chaotic inertial oscillations, circumventing the need for any global reconfiguration of the conformal map τ .

S-II.5 The Essential Reason of Voronoi Failure

Applying our Riemannian control input (20) to a Voronoi partition guarantees local collision avoidance, but it triggers global instability due to the coupling of geometry and inertia. In a Riemannian Voronoi framework, the geodesic boundaries wrap around obstacles, causing the cells to become severely non-convex. Consequently, the target Riemannian centroids undergo discontinuous 'jumps' across the manifold as agents move. When paired with second-order Euler-Lagrange dynamics (3), these jumping targets induce massive proportional error. The agents' inertia prevents them from tracking these discontinuous targets, leading to violent actuator saturation, undissipated kinetic energy, and chaotic inertial oscillations. Our Sectorial partition mathematically eliminates this failure mode by utilizing 1-D virtual partition phases to define the boundaries. This guarantees simply connected subregions and smoothly gliding target centroids,

allowing the Euler-Lagrange dynamics to safely and asymptotically dissipate energy.

S-III Single Integrator

This section discusses the theoretical modifications required when the dynamics are changed to single-integrator, along with the presentation of simulation results.

S-III.1 Agent Dynamics

The simplification of the dynamics facilitates reader comprehension. Meanwhile, this also provides feasibility for extending the system to other dynamics. The dynamics of the i -th agent are presented as

$$\dot{p}_i = u_i, \forall i \in \mathbb{I}_N = \{1, 2, \dots, N\} \quad (\text{S-1})$$

with the control input $u_i \in T_{p_i}S$. Here, $T_{p_i}S$ denotes the tangent space of p_i on S . In light of Lemma 4.2, the control input u_i is designed as

$$u_i = -k_p \nabla_{\eta, p_i} J = -k_p \eta_i^{-1}(p_i) \frac{\partial J}{\partial p_i}, \quad (\text{S-2})$$

where $k_p > 0$ is the gain, and Riemannian metric η_i is the pull-back of $\hat{\eta}_i = (1 + \hat{\sigma}_i)^2 \text{diag}(1, \hat{r}^2)$. This formulation defines u_i as a geometric invariant, where the transformation of η_i naturally compensates for coordinate changes to maintain the consistency across local charts [Kumar *et al.*, 2010].

S-III.2 Theoretical Revision

For Claim 2) of Proposition 3.1, the Euler-Lagrange dynamics should be revised to Riemannian gradient flows. The proposition is that the gradient flows on S and Ξ are topologically conjugate, rendering their asymptotic stability mathematically equivalent via $\mathcal{W}(\mathbf{p}) = \mathcal{V}(\hat{\mathbf{p}})$.

Proof. Since the conformal mapping τ is a global diffeomorphism on the compact manifold S , its Jacobian matrix J_τ is non-singular everywhere, which guarantees a one-to-one correspondence of the equilibria, ensuring that $\dot{\mathbf{p}} = 0 \iff \dot{\hat{\mathbf{p}}} = 0$. By defining $\mathcal{W}(\mathbf{p}) = \mathcal{V}(\tau(\mathbf{p}))$, the bijection preserves the positive definiteness of the energy around the equilibria. Furthermore, the push-forward vector field dictates $\dot{\hat{\mathbf{p}}} = J_\tau \dot{\mathbf{p}}$. Applying the chain rule to the time derivative yields

$$\dot{\mathcal{W}}(\mathbf{p}) = (\nabla_{\mathbf{p}} \mathcal{W})^T \dot{\mathbf{p}} = (\nabla_{\hat{\mathbf{p}}} \mathcal{V})^T J_\tau \cdot J_\tau^{-1} \dot{\hat{\mathbf{p}}} = \dot{\mathcal{V}}(\hat{\mathbf{p}}).$$

Hence, $\dot{\mathcal{V}} < 0$ in Ξ inherently enforces $\dot{\mathcal{W}} < 0$ in S . $\square$

For Claim 3) of Proposition 4.2, the mechanical energy is simplified to the coverage cost because the single-integrator does not include the velocity variable.

Proof. According to Claim 1) of Proposition 4.2, $\eta_i \succ 0$ implies $\eta_i^{-1} \succ 0$ on $\mathcal{F}_i$. Substituting (20) yields $\dot{J} = (\partial J / \partial p_i)^T u_i = -k_p (\partial J / \partial p_i)^T \eta_i^{-1} (\partial J / \partial p_i) \leq 0$. Under Assumption 2.1, $\lim_{p_i \rightarrow \partial \mathcal{F}_i} \hat{\sigma}_i = \infty \implies \lim_{p_i \rightarrow \partial \mathcal{F}_i} J = \infty$. Thus, a feasible initial state $p_i(0) \in \mathcal{F}_i$ strictly ensures $J(t) \leq J(0) < \infty$ for all $t \geq 0$. It confines the trajectory to the compact sublevel set $\Omega_0 = \{\mathbf{p} \mid J(\hat{\psi}, \mathbf{p}) \leq J(0)\} \subset \mathcal{F}_i$, establishing the forward invariance. $\square$

For the single integrator, we only analyze the Riemannian gradient flow. The proof of Proposition 4.3 (Existence and reachability of solutions) is revised to

Proof. The joint configuration space $\mathcal{Q} = S^N \times [0, 2\pi]^N$ is compact as the Cartesian product of compact sets. Since the pull-back metric η_i induces a continuous cost functional J on $\mathcal{Q}$, the Weierstrass Extreme Value Theorem guarantees the existence of a global minimum $(\hat{\psi}^*, \mathbf{p}^*)$ [Boumal, 2023]. For the reachability, excluding the measure-zero cut loci on the Riemannian manifold, the distance function d_l is smooth, ensuring the gradient $\nabla_{\eta} J$ is well-defined [Wolter *et al.*, 1985]. By the Hopf-Rinow Theorem [Petersen, 2006], the compact manifold (S, η_i) is geodesically complete. Concurrently, the metric singularity $\hat{\sigma}_i \rightarrow \infty$ renders the feasible interior strictly forward invariant, guaranteeing the global existence of trajectories $\mathbf{p}(t)$ for all $t \geq 0$. Evaluating the Lyapunov candidate J along the system trajectories yields the dissipation $dJ/dt = -k_p \sum_{i=1}^N \|\nabla_{\eta, p_i} J\|_{\eta_i}^2 \leq 0$. Because the trajectories are bounded within a compact invariant set and $\dot{J} \leq 0$, LaSalle's Invariance Principle dictates the asymptotic convergence to the critical set $\{(\mathbf{p}, \hat{\psi}) \in \mathcal{Q} \mid \nabla_{\eta} J = 0\}$ [Khalil, 2002]. Identifying local minima as the stable equilibria of the gradient flow, the MAS asymptotically stabilizes at the optimal coverage configuration. $\square$

The proof of Theorem 4.2 (Locally exponentially stable of the equilibrium) is revised to

Proof. The partition dynamics $\hat{\psi}$ operates independently of the agent positions p . By Theorem 4.1, $\hat{\psi}(t)$ is exponentially ISS, converging to a steady-state configuration $\hat{\psi}^*$. Substituting $\hat{\psi}^*$ into (3) reduces the agent dynamics to a Riemannian gradient flow $\dot{p}_i = -k_p \nabla_{\eta, p_i} J(\hat{\psi}^*, \mathbf{p})$. Proposition 4.3 dictates $\dot{J} = -k_p \sum_{i=1}^N \|\nabla_{\eta, p_i} J\|_{\eta_i}^2 \leq 0$, ensuring asymptotic convergence to the equilibrium set where $\nabla_{\eta, p_i} J = 0$. The boundary singularity $\eta_i^{-1} \rightarrow 0$ at $\partial \mathcal{F}_i$ guarantees strict forward invariance, rendering the trajectories collision-free. Since $\partial(\nabla_{\eta} J) / \partial \hat{\psi}$ is bounded over the compact manifold, this configuration achieves global asymptotic stability without requiring a timescale separation between k_p and $k_{\hat{\psi}}$. For local exponential stability, we evaluate the Jacobian of the MAS at the equilibrium $(\hat{\psi}^*, \mathbf{p}^*)$. Since $\hat{\psi}$ is independent of p , the system Jacobian is block lower-triangular. Given the exponential stability of $\hat{\psi}^*$ established in Theorem 4.1, the local stability is strictly determined by the spectrum of the Riemannian Hessian operator $\nabla^2 J(\mathbf{p})$ evaluated at the static

configuration $\hat{\psi}^*$. Under Assumption 2.1, the Hessian $\nabla^2 \hat{f}_i$ evaluated in Ξ is given by

$$\nabla^2 \hat{f}_i(\hat{p}_i) = \int_{\hat{E}_i} (f''(d_i) \nabla d_i \otimes \nabla d_i + f'(d_i) \nabla^2 d_i) \hat{\rho}(\hat{q}) d\hat{q}.$$

Since $f', f'' > 0$ (Assumption 2.1) and $\nabla^2 d_i \succ 0$ within the convexity radius of the critical point, it follows that $\nabla^2 \hat{f}_i(\hat{p}_i^*) \succ 0$. The conformal pull-back $J = \hat{f} \circ \tau$ induces the congruence transformation $\nabla^2 J(p^*) = (J_\tau)^T \nabla^2 \hat{f}(\hat{p}^*) J_\tau$. Because the diffeomorphism τ renders J_τ non-singular, Sylvester's Law of Inertia preserves positive definiteness, yielding $\nabla^2 J(p^*) \succ 0$ [Petersen, 2006]. Since the established equilibrium p_i^* satisfies both the first-order necessary condition ($\nabla_\eta J = 0$) and the second-order sufficient condition ($\nabla^2 J \succ 0$), it coincides exactly with the optimal Riemannian centroid. The linearized system matrix $-k_p \nabla^2 J(p^*)$ possesses strictly negative eigenvalues, thereby establishing local exponential stability at the optimal centroid. $\square$

The proof of Theorem 4.3 (optimal approximation) is revised to

Proof. Let $\mathcal{X} = (\hat{\psi}, \mathbf{p}) \in \mathbb{T}^N \times S^N$, where $\mathbb{T}^N$ denotes the N -dimensional torus parameterizing the periodic partition phases. By the triangle inequality, the total cost deviation comprises three discretization errors and a finite-horizon residual ϵ_{conv} : $|J(\mathcal{X}^*) - J(\mathcal{X}^A)| \leq \Delta J_1 + \Delta J_2 + \Delta J_3 + \epsilon_{\text{conv}}$, where ϵ_{conv} represents the convergence residual due to the finite termination horizon T_ϵ . The term $\Delta J_1 = |J(\hat{\psi}^*, \mathbf{p}^*) - J(\hat{\psi}^*, \mathbf{p}^k)|$ quantifies the steady-state discrepancy between the continuous trajectory $\mathbf{p}(t)$ and the discrete sequence $\mathbf{p}^k$, which is updated via Riemannian retraction [Boumal, 2023]. The geodesic completeness of S (Proposition 4.2) renders the discrete update globally well-defined. Applying the second-order covariant Taylor expansion, the cost variation is dominated by the Riemannian Hessian $\nabla^2 J$. Invoking Assumption 2.1 ($f', f'' > 0$), the compactness of S ensures a finite upper bound $L_H = \sup_{p \in S} \|\nabla^2 J(p)\|_l$, where $\|\cdot\|_l$ denotes the operator norm induced by the length metric $d_l(\cdot)$. L_H explicitly incorporates the curvature of S and the performance supremum $\bar{f}'' = \sup f''$, one gets

$$|J(\hat{\psi}^*, p^*) - J(\hat{\psi}^*, p^k)| \leq \frac{1}{2} L_H \|\nabla_\eta J\|_{l, \max}^2 (\Delta t)^2,$$

where the zeroth and first-order terms cancel as the discrete retraction is aligned with the Riemannian gradient flow. The term $\Delta J_2 = |J(\hat{\psi}^*, \mathbf{p}^k) - J(\hat{\psi}^k, \mathbf{p}^k)|$ stems from quantizing the configuration torus $\mathbb{T}^N$ into K^* intervals, which induces $d_{\mathbb{T}}(\hat{\psi}^*, \hat{\psi}^k) \leq 2\pi/K^*$. By invoking Assumption 2.1 ($f' > 0$) and applying the Stokes' Theorem [Petersen, 2006] to the boundary $\partial \hat{E}_i$, the intrinsic gradient is evaluated as $\nabla_{\hat{\psi}_i} J = [f(d_l(p_i, q)) - f(d_l(p_{i-1}, q))] \hat{\omega}(\hat{\psi}_i)$, where $\hat{\omega}(\hat{\psi}_i)$ denotes the marginal density at the i -th partition bar. Consequently, the quantization error is bounded by

$$|J(\hat{\psi}^*, p^k) - J(\hat{\psi}^k, p^k)| \leq L_{\hat{\psi}}(2\pi/K^*),$$

where $L_{\hat{\psi}} = \sup_{\hat{\psi}} \|\nabla_{\hat{\psi}} J\|_l$ is constrained by $\bar{f}' = \sup f'$ and the pull-back volume form across $\partial \hat{E}_i$.

The term $\Delta J_3 = |J(\mathcal{X}^k) - J(\mathcal{X}^A)|$ reflects the domain perturbation induced by the sequence factor β . The rerouting of partition bars from B_i to $\hat{\Gamma}_i$ induces a domain perturbation $\Delta \mathcal{M}_i \subset S$. By invoking Stokes' Theorem [Petersen, 2006], the cost deviation is estimated by

$$|J(\mathcal{X}^k) - J(\mathcal{X}^A)| = \left| \sum_{i=1}^N \int_{\Delta \mathcal{M}_i} f(d_l(p_i, q)) \rho(q) dV_i \right| \leq L_g C_\delta \beta,$$

where $L_g = \sup f(d_l)$ is the performance bound on S . In light of Lemma 4.1, the workload deviation $|\Delta \hat{m}_i| \leq C_\delta \beta$ linearly maps to the cost penalty via the conformal factor λ^2 of the pull-back volume form $dV_i = \det(\tau^* \hat{\eta}_0)^{1/2} dx$. Finally, since $\nabla^2 J \succ 0$ at the equilibrium (Theorem 4.2), the objective functional exhibits geodesic strong convexity [Boumal, 2023]. This quadratic growth condition translates the total cost deviation into a strict state deviation bound [Bonnabel, 2013]

$$d_l(\mathcal{X}^A, \mathcal{X}^*) \leq \sqrt{2|J(\mathcal{X}^A) - J(\mathcal{X}^*)|/\lambda_{\min}(\nabla^2 J)},$$

where $\lambda_{\min}(\nabla^2 J)$ is the minimum eigenvalue of the Riemannian Hessian. Consequently, appropriately tuning the precision parameters Δt , K^* , and β compresses the practical optimality gap arbitrarily close to zero. $\square$

S-III.3 Simulations of single integrator

Compared with EL dynamics, the single-integrator model eliminates robot inertia, enabling motion at arbitrary angles. Using the mapping and partition designed in the original work, the experimental phenomena under single-integrator dynamics are obtained as follows, with the simulation environment maintained unchanged.

As illustrated in Fig. S-1 and Fig. S-2, the single-integrator agents navigate the ball world Ξ and the poly-annulus S to achieve optimal coverage. Unlike the Euler-Lagrange system, the absence of inertia permits the agents to execute sharp, omnidirectional trajectory adjustments, enabling more direct paths toward their optimal Riemannian centroids. The quantitative performance indices in Fig. S-3 confirm this efficiency, as the individual workloads m_i rapidly converge to an equitable equilibrium, the partition phases ψ_i stabilize, and the control inputs u_i strictly dissipate to zero. Furthermore, the framework's robustness is validated through a simulated hardware failure at $t = 200$. As shown in Fig. S-4 and Fig. S-5, the remaining active agents autonomously execute a self-healing response by expanding their sectorial boundaries to subsume the abandoned regions. Fig. S-6 demonstrates that the system accommodates

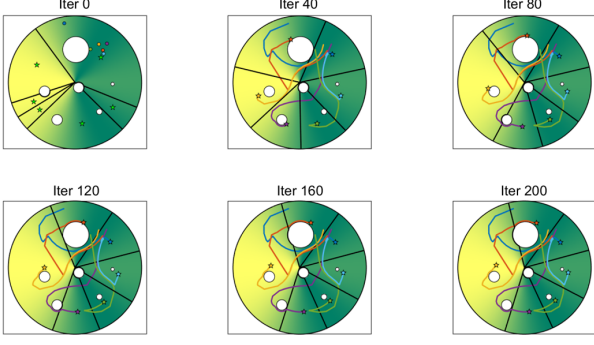


Fig. S-1. Temporal evolution of the MAS in the ball world Ξ . Black radial lines denote the safe sectorial partitions, circles represent agent positions, and stars indicate the optimal Riemannian centroids. The single integrator dynamics exhibit rotation in arbitrary directions.

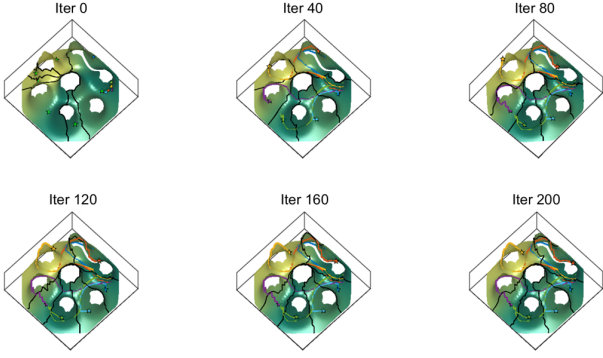


Fig. S-2. Corresponding coverage evolution in poly-annulus S . Through the inverse conformal mapping τ^{-1} , the linear sectors in Ξ are transformed into curvilinear regions in S that naturally conform to the irregular obstacle boundaries while maintaining simply connected topologies.

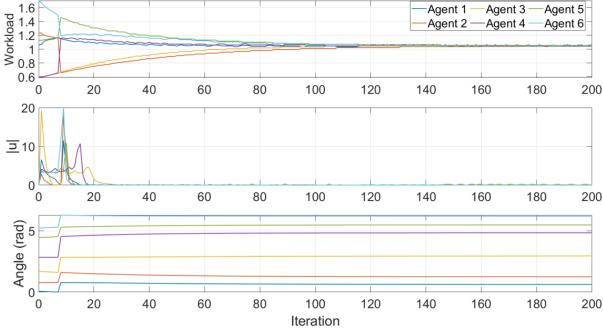


Fig. S-3. Evolution of performance indices: (Top) Convergence of agent workloads m_i to the equilibrium; (Middle) Asymptotic decay of control inputs u_i ; (Bottom) Stabilization of partition phases ψ_i .

this severe disruption with a swift transient response, automatically compensating for the lost agents and establishing a new, elevated workload equilibrium without requiring a re-computation of the conformal mapping. Ultimately, these results confirm that the proposed coverage framework remains stable and robust, possessing the intrinsic flexibility to be extended to varying robot dynamics.

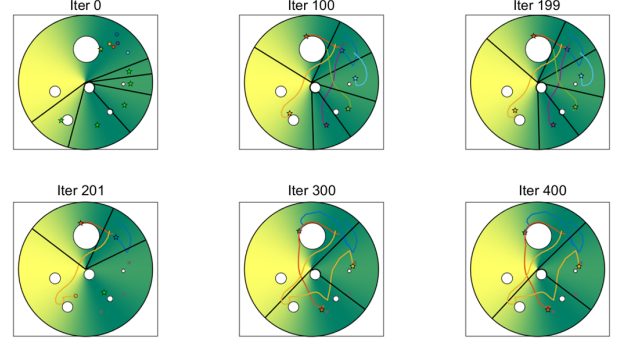


Fig. S-4. Self-healing response in Ξ under agent failure at $t = 200$. The active agents autonomously expand their sectorial angles to cover the regions vacated by the failed agents.

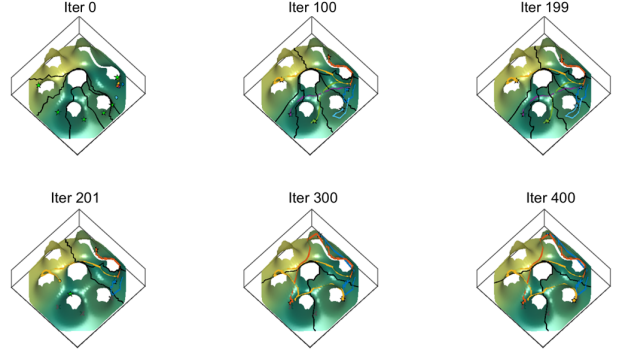


Fig. S-5. Reconfiguration of the MAS in S following agent failure. The remaining agents redistribute the global workload without requiring global re-mapping.

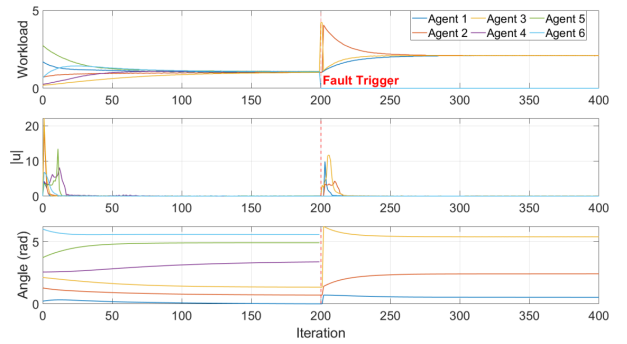


Fig. S-6. Performance indices during the failure recovery process. The dashed line at $t = 200$ marks the fault occurrence. The system exhibits a rapid transient response, where workloads m_i re-converge to a new, higher equilibrium, verifying the automatic rate compensation capability.